

CF07 CMAN

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a) Dense and sparse compute/memory

For a dense $N \times N$ MVM, there are N^2 multiply-accumulate operations. Counting one multiply and one add per MAC gives:

$$F_{\text{dense}} = 2N^2 = 2(512)^2 = 524,288 \text{ FLOPs}$$

Dense matrix memory is:

$$B_{\text{dense}} = 4N^2 = 4(512)^2 = 1,048,576 \text{ bytes}$$

With sparsity s , the nonzero fraction is $1 - s$, so the number of nonzeros is:

$$\text{nnz} = (1 - s)N^2$$

Sparse compute is therefore:

$$F_{\text{sparse}} = 2(1 - s)N^2 = 2(1 - s)(512)^2 = 524,288(1 - s) \text{ FLOPs}$$

Sparse memory is one 4-byte value and one 4-byte column index per nonzero, plus $N + 1$ row pointers:

$$B_{\text{sparse}} = (4 + 4)(1 - s)N^2 + 4(N + 1)$$

$$B_{\text{sparse}} = 8(1 - s)(512)^2 + 4(513)$$

$$B_{\text{sparse}} = 2,097,152(1 - s) + 2,052 \text{ bytes}$$

b) FLOPs speedup

The theoretical FLOPs speedup is the dense FLOPs divided by the sparse FLOPs:

$$S_{\text{FLOPs}} = \frac{F_{\text{dense}}}{F_{\text{sparse}}} = \frac{2N^2}{2(1 - s)N^2} = \frac{1}{1 - s}$$

For a $2\times$ FLOPs speedup:

$$\begin{aligned} \frac{1}{1 - s} &= 2 \\ 1 &= 2(1 - s) \end{aligned}$$

$$1 = 2 - 2s$$

$$2s = 1$$

$$s = 0.5$$

So, sparse MVM has a theoretical $2\times$ FLOPs speedup at 50% sparsity.

c) Memory breakeven

The memory breakeven occurs when sparse memory equals dense memory:

$$\begin{aligned} B_{\text{sparse}} &= B_{\text{dense}} \\ 8(1-s)N^2 + 4(N+1) &= 4N^2 \end{aligned}$$

Divide by 4:

$$\begin{aligned} 2(1-s)N^2 + (N+1) &= N^2 \\ 2(1-s)N^2 &= N^2 - N - 1 \\ 1-s &= \frac{N^2 - N - 1}{2N^2} \\ s &= 1 - \frac{N^2 - N - 1}{2N^2} \\ s &= \frac{N^2 + N + 1}{2N^2} \end{aligned}$$

For $N = 512$:

$$s = \frac{512^2 + 512 + 1}{2(512)^2} = \frac{262,657}{524,288} \approx 0.50098$$

So, the sparse format uses less memory above about 50.1% sparsity - would be 50% without the CSR row pointer overhead.

d) Memory-bound speedup at $s = 0.9$

At $s = 0.9$, the nonzero fraction is $1 - s = 0.1$. The dense memory traffic is:

$$B_{\text{dense}} = 1,048,576 \text{ bytes}$$

The sparse memory traffic is:

$$B_{\text{sparse}} = 2,097,152(0.1) + 2,052$$

$$B_{\text{sparse}} = 211,767.2 \text{ bytes}$$

For a memory-bandwidth-limited system with bandwidth 320 GB/s:

$$T_{\text{dense}} = \frac{B_{\text{dense}}}{320 \times 10^9} = \frac{1,048,576}{320 \times 10^9} = 3.2768 \times 10^{-6} \text{ s}$$

$$T_{\text{sparse}} = \frac{B_{\text{sparse}}}{320 \times 10^9} = \frac{211,767.2}{320 \times 10^9} = 6.6177 \times 10^{-7} \text{ s}$$

Therefore, the end-to-end memory-bound speedup is:

$$S_{\text{end-to-end}} = \frac{T_{\text{dense}}}{T_{\text{sparse}}} = \frac{3.2768 \times 10^{-6}}{6.6177 \times 10^{-7}} \approx 4.95$$

So, on a memory-bound system the speedup would be limited to about 4.95 $\times$.